Carbon Removal Verification Registry

Project Design Report

Course ITCS514 Software Project Management **Group** 1 **Term** 1, Y2026/27
Client Julian Ferris, Head of Registry Integrity

1 Problem background

1%

A carbon removal registry certifies that a tonne of carbon dioxide has genuinely been removed from the atmosphere — through reforestation, soil carbon or similar — and issues a credit a corporate buyer can purchase against its emissions. The credit is only worth what the verification behind it is worth.

Today that verification is manual and trust-based: developers email documents and photographs, staff and verifiers review them by hand, and the evidence is filed away. Three consequences follow, and all three are now public.

Land boundaries are never checked against each other. Nothing prevents two developers registering overlapping parcels, so the same hectare can be sold twice. Boundaries are held as PDFs and scanned maps, not as geographic data, so no systematic screening is possible.

Nothing is monitored after approval. Once a project is certified, no one checks whether the forest is still standing. A credit issued in 2026 says nothing about the land in 2027.

Evidence cannot be produced on demand. Answering “why was this credit issued?” takes weeks of searching email and shared drives, and nothing proves the record was not altered afterwards.

Public reporting of these weaknesses has had a direct commercial effect: **three major corporate buyers have paused approximately USD 2 million in pending purchases** until the registry can show auditable proof of credit integrity. The largest represents roughly 30% of issuance volume. The registry's product is credibility, and credibility is what has failed.

Intended users

User	What they need from the system
Project developer	Register a project, draw its boundary, upload evidence, and respond to monitoring alerts — often from the field, on a phone, with intermittent connectivity.
Independent verifier	Work a queue of assigned projects, see the evidence and the automatic checks in one place, and record a signed verification decision.
Registry staff / admin	Approve or reject verified projects under a maker-checker control, manage users and roles, and answer questions about any issued credit.
Corporate buyer	Trace any credit back to the evidence and the verifier behind it, and see whether the project is still performing.
Sponsor and board	Evidence that the integrity problem has been addressed, in time for the board demonstration and the buyer review.

Scope of the proposed solution

We propose a **centralised web application** that brings registration, geospatial validation, independent verification, buyer access and auditability into one system. The application is responsive and works down to phone width, so a developer can draw a boundary in the field without a separate mobile app.

In scope: standardised project registration with evidence capture; digital boundaries with automatic overlap detection before issuance; a structured verification workflow with maker-checker approval; controlled buyer access to evidence and credit traceability; role-based access control and a tamper-evident audit trail.

Out of scope: replacing physical site verification, automated additionality decisions, migration of all existing projects, cross-registry reconciliation, advanced internal dashboards, full workflow automation, and any blockchain or distributed-ledger component. Blockchain was considered and rejected: it secures a ledger but does nothing to prove the underlying carbon evidence is true, which is the actual problem.

2 Expected benefits

1%

The current process is slow, unverifiable after the fact, and cannot detect double-selling at all. The proposed system changes each of those.

- **Double-selling becomes detectable before a credit is issued, not after.** Every boundary is stored as geographic data and screened against every other registered boundary automatically. Today this check does not happen at all; under the new system an overlap blocks issuance and raises a flag. *Benefits: registry, buyers.*
- **Projects are monitored continuously instead of once.** Satellite land-cover comparison runs against registered boundaries after approval, so canopy loss raises an alert rather than going unnoticed until the next manual review. *Benefits: buyers, registry.*
- **Any credit can be traced to its evidence in seconds rather than weeks.** A buyer opens the credit and sees the project, the boundary, the verifier, the sign-off date and the original evidence files. Today the same question takes days of searching email and shared drives. *Benefits: buyers, registry staff.*
- **Evidence cannot be altered after submission.** Files are written once and never edited or deleted, and every critical action is recorded in an append-only ledger where each entry seals the one before it. Silent changes become visible. *Benefits: buyers, auditors, the board.*
- **Verification becomes consistent and reviewable.** Verifiers work from the same structured checklist with the automatic checks already run, instead of each reviewing documents in their own way. *Benefits: verifiers, registry.*
- **Separation of duties is enforced by the system.** The person who submits cannot be the person who approves — a maker-checker control applied in software rather than by convention. *Benefits: registry integrity, auditors.*
- **The commercial outcome the client is actually buying:** the evidence package needed to restart the paused purchases. The target is the resumption of the approximately USD 2 million currently on hold, supported by a working pilot that buyers can inspect themselves. *Benefits: the registry's revenue and reputation.*

A note on what we are not claiming. We cannot claim the system makes carbon claims true — only that it makes the evidence behind them complete, traceable and tamper-evident, and that it removes a category of error (overlapping land) that the manual process cannot catch. Physical verification remains a human activity and stays out of scope.

3 Proposed features

3%

Five features form the web application we will build. Each is traced to the charter's functional requirements and to a branch of the project work breakdown structure.

Project registration & evidence

FR-01 · FR-02 · WBS 4.0 F1

WHAT — A guided multi-step form covering project identity, methodology, crediting period, area and baseline, then evidence upload — land title, survey PDF, geo-tagged photographs. Files are sealed on submission and cannot be edited or replaced. Status is tracked on a dashboard.

WHO — Project developers submit; registry staff view all submissions.

WHY — Intake today is email and spreadsheets, so submissions are inconsistent and evidence scattered. Nothing can be checked automatically if it arrives as an attachment, so standardised intake is the precondition for every later feature.

Geospatial boundary & overlap detection

FR-03 · FR-04 · WBS 5.0 F2

WHAT — Boundaries are captured by uploading a GPS or geospatial file, or by drawing the polygon on a map — including on a phone in the field. Each is stored as geographic data and screened against every registered boundary. An intersection names the conflicting project, shows the overlapping area, and blocks issuance until resolved.

WHO — Project developers draw and submit; verifiers and registry staff see the screening result.

WHY — This is the client's highest priority and the feature that addresses double-selling, the most damaging of the three failures. No equivalent check exists today.

Verification workflow

Charter §5.3 · FR-10 · WBS 7.0 F3

WHAT — A work queue of assigned projects with due dates and overdue flags. Opening one gives the verifier the evidence, the automatic check results and a structured checklist — land title matches applicant, no boundary overlap, baseline complete, methodology correctly applied. The decision is then approved or rejected by a different member of registry staff.

WHO — Independent verifiers review; registry staff perform the second approval.

WHY — Verification today varies by individual and leaves no structured record. A shared checklist makes decisions consistent and makes separation of duties enforceable rather than conventional. The charter carries this as an in-scope boundary (§5.3, “structured verification workflow”) rather than a numbered requirement; the maker-checker control is FR-10.

Public registry & buyer transparency

FR-08 · WBS 8.0 F4

WHAT	A searchable registry of issued credits. Selecting one shows its full traceability — serial, vintage, project, verifier, sign-off date, status and latest monitoring result — and gives controlled access to the underlying evidence, filtered by the viewer's role.
WHO	Corporate buyers and the public; registry staff control what each buyer may see.
WHY	This is the feature the paused buyers will actually judge the registry on. It converts internal rigour into something an outsider can verify without asking anyone.

Security & auditability

FR-07 · FR-09 · FR-10 · WBS 9.0 F5

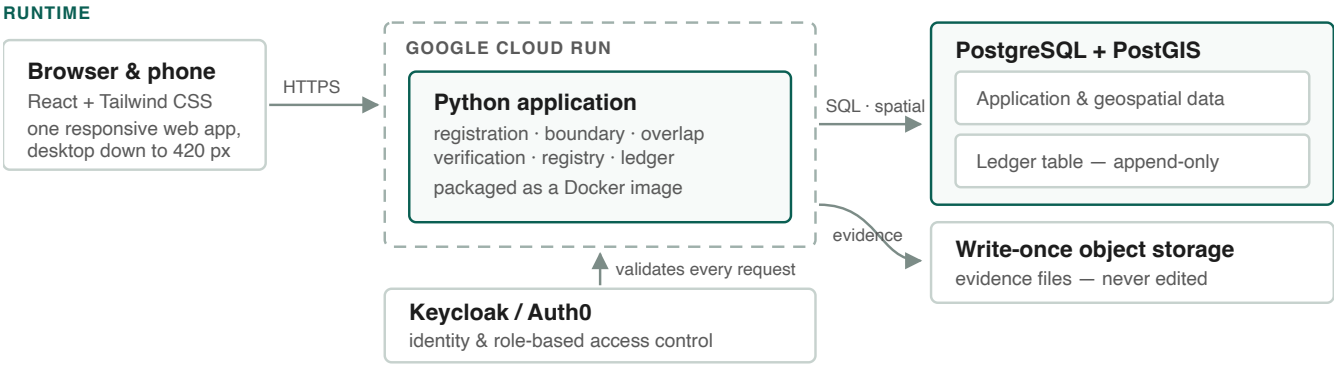
WHAT	Authentication and role-based access control across the five user roles the charter names, with separation of duties enforced in software. Every critical action — submission, boundary change, verification decision, approval, issuance — is written to an append-only ledger in which each entry carries a cryptographic hash of the one before it, so any later alteration breaks the chain and becomes visible. Evidence files are held in write-once storage.
WHO	Applies to every user; the ledger is read by registry staff, auditors and buyers.
WHY	Without this the other four features are merely convenient. It is what lets the registry answer "prove this was not changed afterwards", which is the question the buyers are really asking.

Designed, but not built this term. Two features of the full MVP are in the solution design and the architecture, but outside this build.

Satellite monitoring (FR-05 · WBS 6.0) — continuous land-cover change detection against registered boundaries. It needs a commercial satellite imagery subscription the team cannot procure. The interface reserves its place: the dashboard shows monitoring alerts and the data model stores monitoring records.

Inter-registry flagging hook (FR-11 · WBS 9.4) — an internal endpoint for future duplicate checks against other registries. Nothing external is connected, so there is no counterpart to build against; the endpoint and its data contract are designed so the integration can be added later without rework.

Platform





Runtime and delivery. One Python service behind a single login check, reading and writing one database and one evidence store. Every release reaches production the same way — built, tested and deployed from GitHub, never copied by hand.

Layer	Technology
Front end	React with Tailwind CSS — one responsive web application, usable from desktop down to phone width
Back end	Python
Data	PostgreSQL with PostGIS for application and geospatial data, including the append-only ledger table; object storage for write-once evidence files
Identity	Keycloak / Auth0 — an existing identity service rather than hand-written login
Delivery	Docker containers, GitHub Actions for CI/CD, deployed to Google Cloud Run

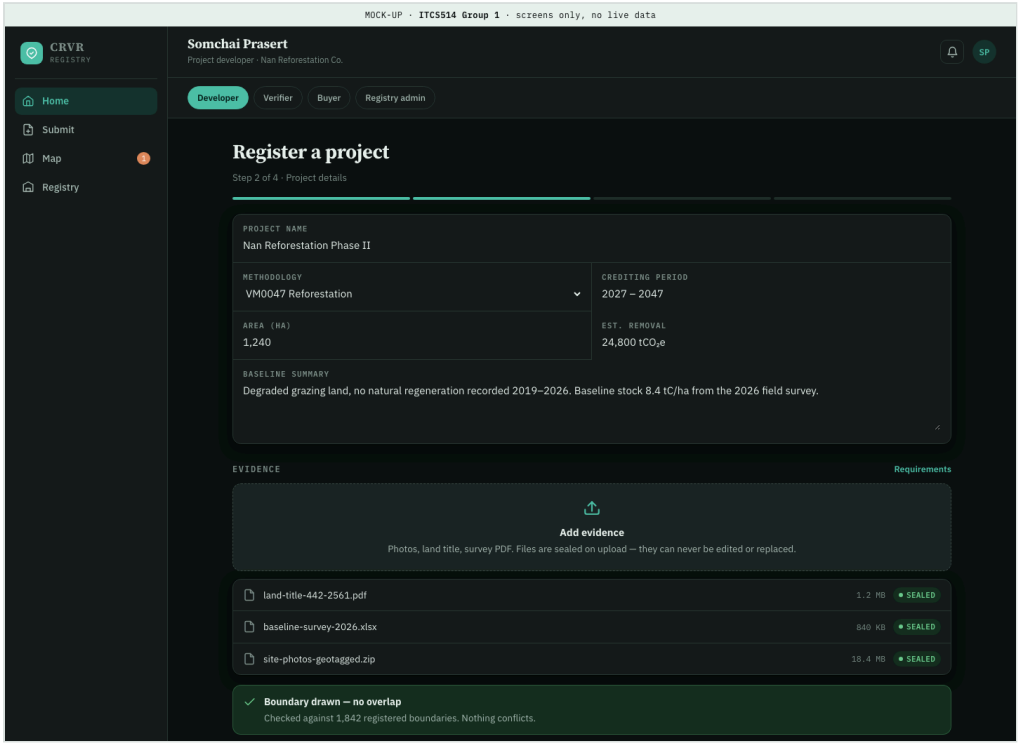
PostGIS is what makes F2 feasible: polygon intersection is a database operation rather than something we implement ourselves, which is why the highest-risk feature is affordable within one term.

4 UI design

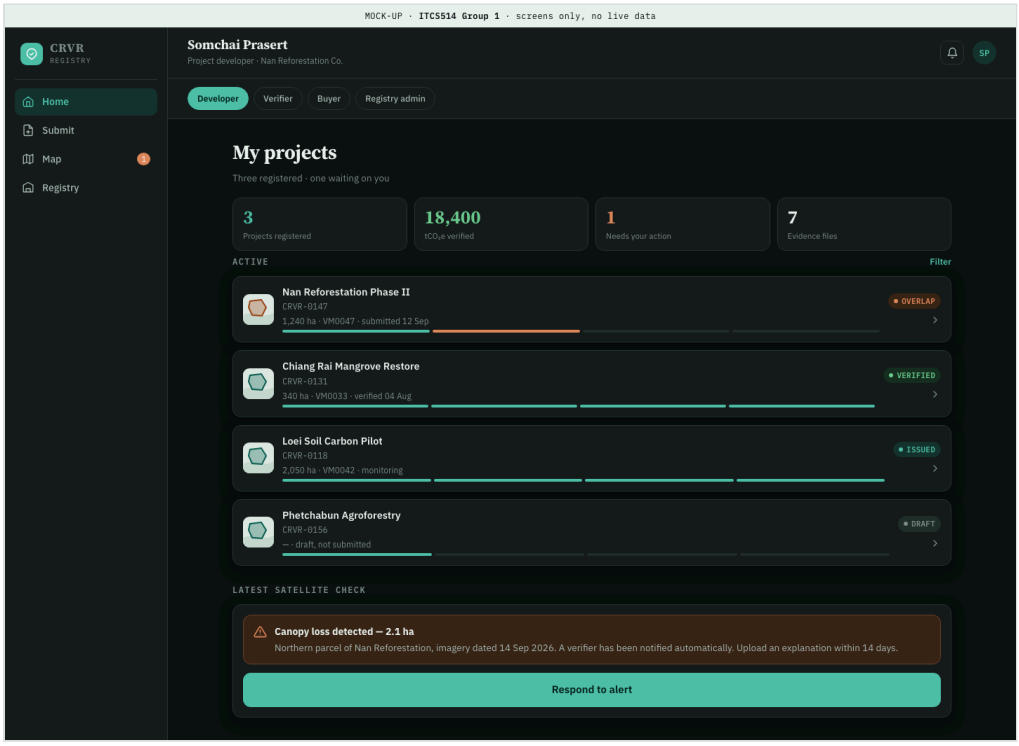
3%

Interactive mock-ups covering the main user flow for every feature. Screens show realistic labels, fields and example content; all data shown is illustrative.

F1 Project registration & evidence

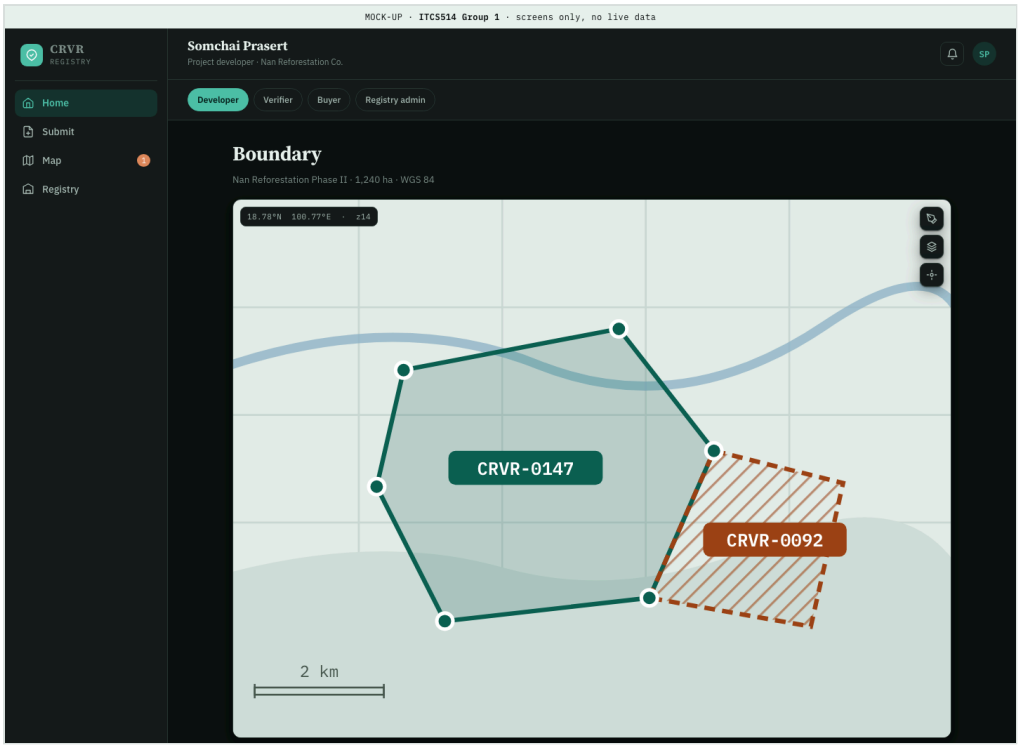


Register a project — step 2 of 4. Project identity and methodology, then evidence upload. Files are marked *sealed* once submitted, signalling that they can never be edited or replaced. The green banner reports the result of the automatic boundary check inline, so the developer learns about a conflict before submitting rather than weeks later.



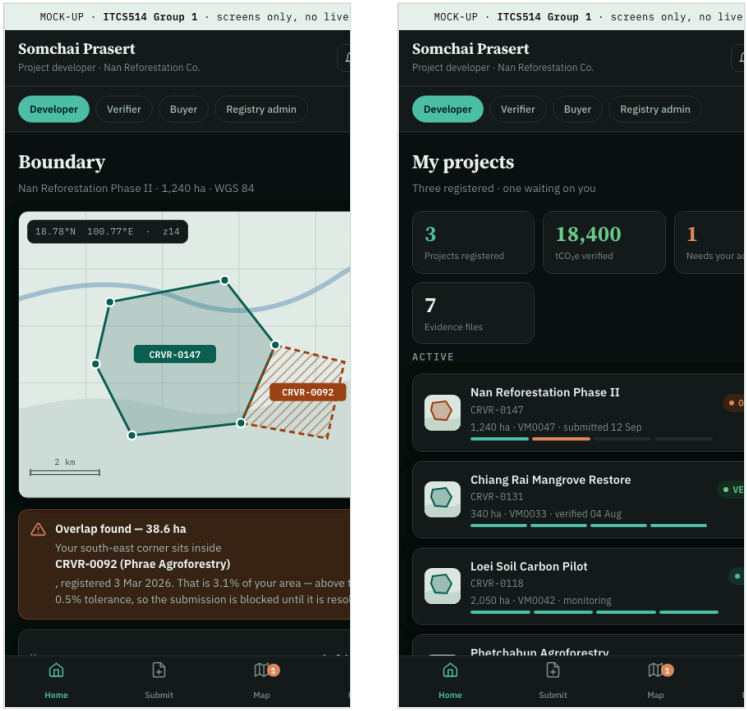
Developer dashboard. Submission status per project, with the one project needing action called out. The amber alert reserves space for monitoring — designed, but not built this term.

F2 Geospatial boundary & overlap detection



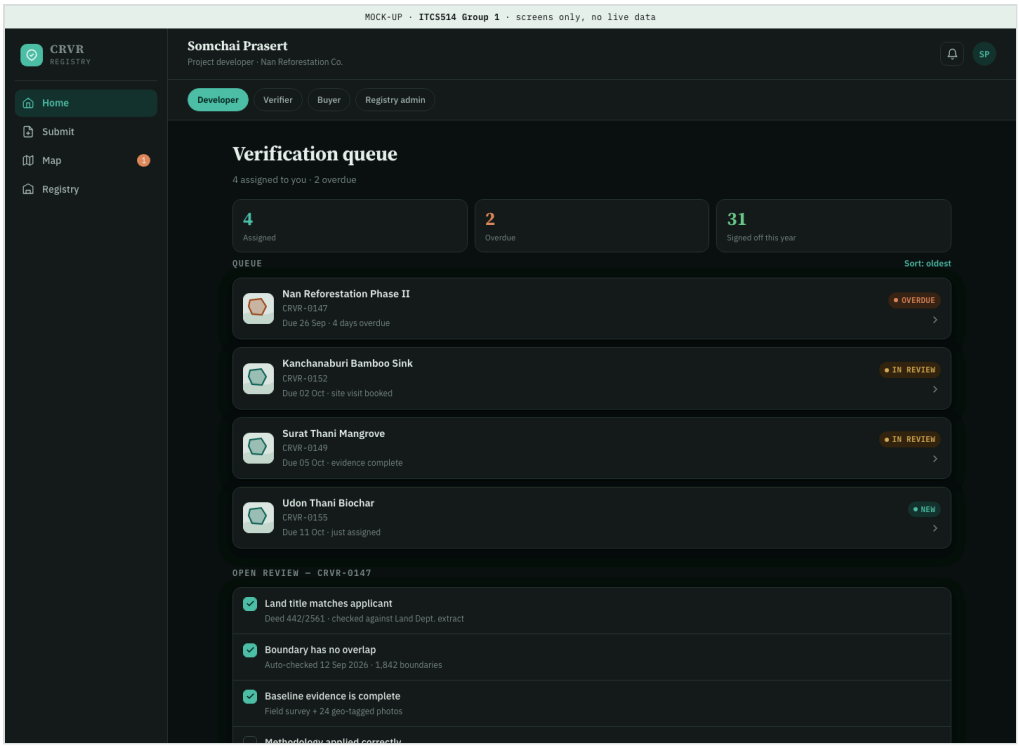
Boundary editor with an overlap detected. The developer's polygon (CRVR-0147, solid) is drawn against the registered boundaries around it; the hatched region is the conflicting project CRVR-0092. Vertices are draggable, coordinates and zoom are shown top-left, and the scale bar supports field use. This single screen is the double-selling control — the thing the current process cannot do at all.

F2 Same screens at phone width



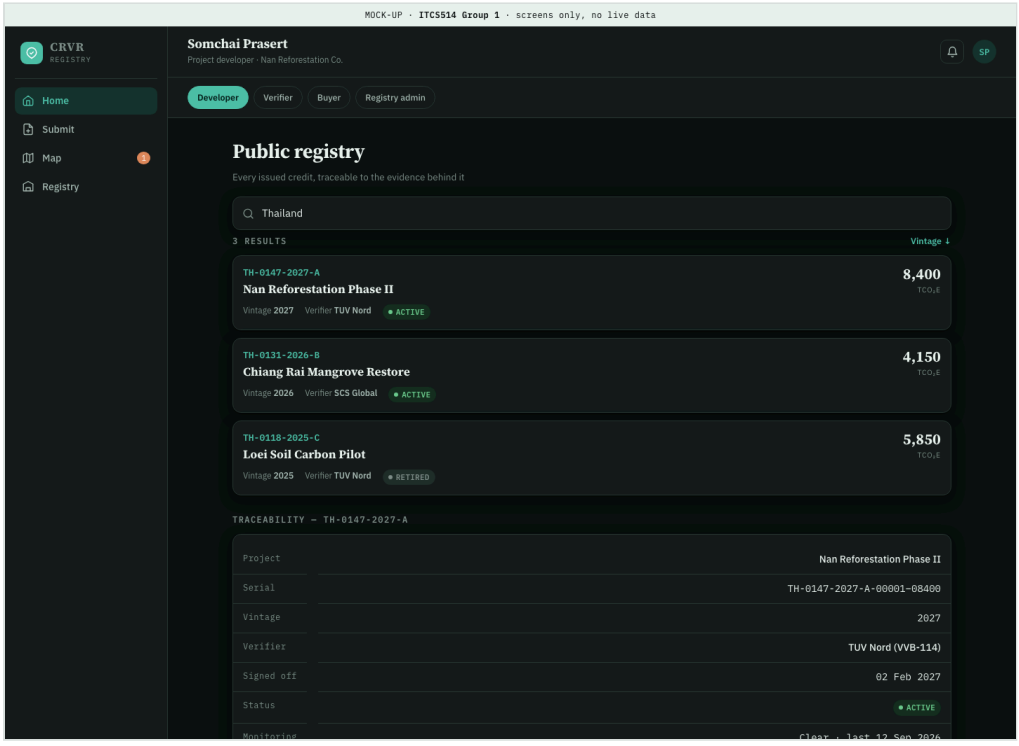
Responsive behaviour. The same web application at 420 px. The sidebar collapses to a bottom navigation bar and the map fills the viewport, so a developer can capture a boundary standing in the field. There is no separate mobile application to build or maintain.

F3 Verification workflow



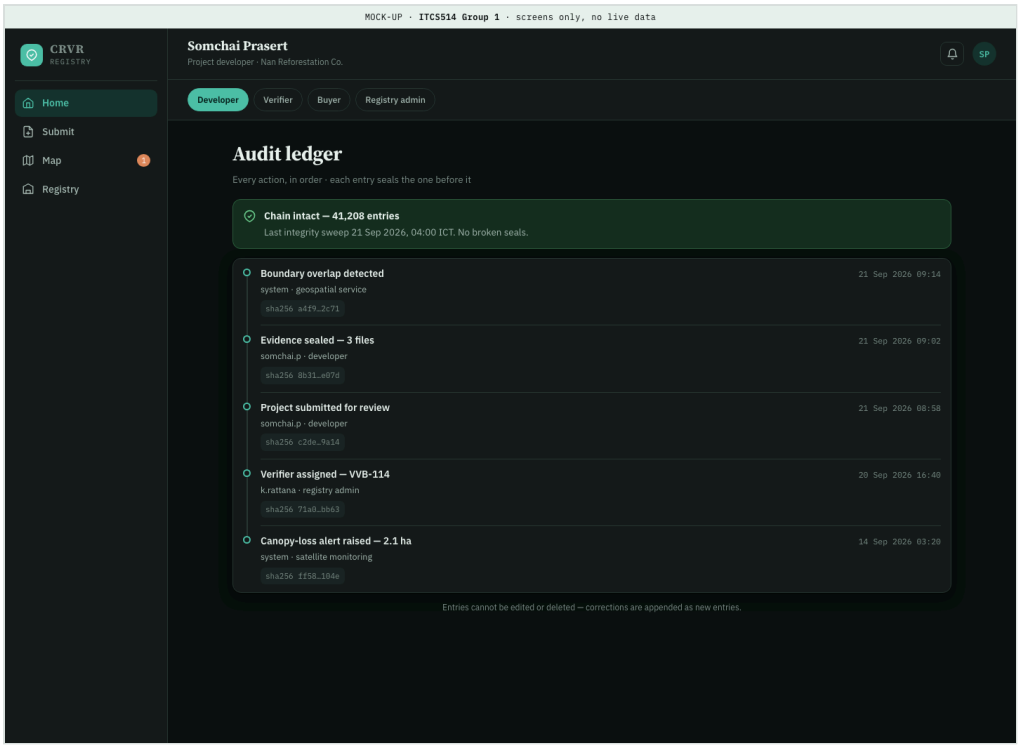
Verifier queue and open review. Assigned projects sorted oldest first with overdue items flagged. Below, the structured checklist for the open review — each item shows what was checked and against what source, including the automatic boundary result. The decision recorded here is approved separately by registry staff, which is how separation of duties is enforced.

F4 Public registry & buyer transparency



Public registry with the traceability panel open. A buyer searches issued credits, then opens one to see serial, vintage, verifier, sign-off date, status and the latest monitoring result. This is the view the paused buyers will be shown at the review.

F5 Security & auditability



Audit ledger. Every critical action in sequence with actor, timestamp and the hash chaining each entry to the previous one. Role switching is visible along the top of every screen — the mock-up carries Developer, Verifier, Buyer and Registry admin views, which is how role-based access control appears to the user.

5 Team responsibilities

2%

Contribution to this design report

Member	Student ID	Role	Contribution to this report
Zay Yar Phyo	6838857	Full-stack	UI design section; all screen mock-ups, flows and example content; responsive behaviour
Nyi Htut Zaw	g6838859	Full-stack	Proposed features section; traceability of features to charter requirements and WBS branches; platform decisions
Pyae Phyo Hein	g6937059	Full-stack	Team responsibilities section; assembly and submission of the final PDF
Jingqing [to confirm]	[to confirm]	Full-stack	Problem background; stakeholder analysis; scope boundaries
Put Thitisawat	6938719	Full-stack	Expected benefits section; review pass for consistency against the project charter, WBS and architecture

Expected contribution for implementation

Every member works full-stack and owns one feature end to end — its data model, back-end logic, interface and tests. Each also supports one other feature, and the cross-cutting work (shared component library, deployment pipeline) is split between members rather than given to a specialist.

> **Why five people here and twelve in the project plan.** The project plan prices the full fourteen-week MVP at 325 person-days across twelve people in seven roles — that is the staffing the client would fund. This report covers what our five-person team will build within the course — five of the seven MVP features, one owned by each member, with everyone working full-stack. The two teams are not in conflict: one is the costed plan, the other is the subset we implement.

Member	Role	Owns	Also responsible for
Zay Yar Phyo	Full-stack	**F4** Public registry & buyer transparency	Shared UI component library and responsive layout; supports F1
Nyi Htut Zaw	Full-stack	**F2** Geospatial boundary & overlap detection	PostGIS schema and spatial queries; supports F1; supports Put Thitisawat on deployment and the CI/CD pipeline
Pyae Phyo Hein	Full-stack	**F3** Verification workflow	Supports F2 overlap resolution; maker-checker logic
Jingqing [to confirm]	Full-stack	**F1** Project registration & evidence	Supports F3; shared data model
Put Thitisawat	Full-stack	**F5** Security & auditability	Authentication and RBAC integration; Docker, GitHub Actions and Cloud Run deployment

F2 carries the highest technical risk — it is the client's first priority and the only feature with no equivalent in the current process — so it is owned by one developer with a second providing support, rather than being split.